KidWind 4: Tell what the wind taught us

Goal: I can explain what we changed, what happened, and the proof.

35 minutes | 7 steps | Engineering and Design | Playbook A meeting 13, grade 5 path; Playbook B meeting 14, grade 5 path

Ask | Plan | Build | Test | **Improve**

You are done when

Say one thing the wind taught you.

You will use

- One teacher-run test station, built and piloted first from parts on hand
- One guarded fan at one fixed setting, with tape marks and a ruler
- A smooth, blunt shaft turning in a larger straw or tube, with a cork or foam hub
- String on a fixed spool, one small cup and the same load pieces every trial
- Cardstock or index cards, reusable blade supports, tape and student scissors
- A marked blade-angle example, timer, shallow catch tray, team folders and record sheets
- Printed practice data table for the no-fan paper path
- Three picture prompt cards: what we changed, what happened, our proof

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 2 min

- Quick recap: what did our blades do?

2 Step 2: Find your records — 1 min

- Get your folder and your best set.

3 Step 3: Final test — 8 min

- Run your best set for one 10 second trial.
- Record the centimeters lifted.

4 Step 4: Plan your talk — 12 min

- Plan a 30 to 45 second talk.
- Say what we changed, what happened, and our proof in centimeters.

5 Step 5: Share your talk — 7 min

- Give your talk to another team.
- Listen and ask one question.

6 Step 6: Reset the station — 3 min

- Fan off first.
- Rotor sets go flat in your team folder.
- Cup and load pieces go back in the station tray.

7 Step 7: Show your result — 2 min

- Say one thing the wind taught you.

What to turn in

A 30 to 45 second talk with a centimeter result as proof.

No computer? Another way to do it

No-fan paper path: follow the paper steps and the labeled practice data below. It uses the same measurement but does not claim you built or tested a windmill.

Finished early? Next challenge

Draw a bar chart of your three trials from last time.

Stay safe

Only the teacher runs the fan and prepares any holes with pointed tools. Stay out of the rotor's spin path. Fan off before anyone touches blades, the cup or the string. Blades are paper or cardstock only. Shaft ends are blunt.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.

No-fan paper path

Use these paper steps instead when your teacher chooses the no-fan path.

Goal: I can explain how wind can lift a load and use sample results to plan a fair test.

1. Plan a 30 to 45 second talk.
2. Say the change you propose and the practice evidence.
3. Say what a real test must check.
4. Call the numbers practice data.

Keep: A talk or written card that calls the numbers practice data.

Practice data created for this lesson. These are not student measurements or evidence about a real blade design.

Unit: Centimeters lifted in 10 seconds with one unchanged cup and load

Practice data created for this lesson. These are not student measurements or evidence about a real blade design.

Practice design	Only stated change	Trial 1	Trial 2	Trial 3
A	Baseline	5	10	5
B	Blade angle only	10	15	10
C	Blade size only, compared with A	10	10	15